Artificial Intelligence Technology and Application

Deep Learning Lab Guide

Student Version

HUAWEI

HUAWEI TECHNOLOGIES CO., LTD.

About This Document

Description

This guide introduces the following five exercises:

- Exercise 1: MindSpore basics, which describes the basic syntax and common modules of MindSpore.
- Exercise 2: Handwritten character recognition, in which the MindSpore framework is used to recognize handwritten characters.
- Exercise 3: MobileNetV2 image classification, which mainly introduces the classification of flower images using the lightweight network MobileNetV2.
- Exercise 4: ResNet-50 image classification exercise, which mainly introduces the classification of flower images using the ResNet-50 model.
- Exercise 5: TextCNN sentiment analysis, which mainly introduces sentiment analysis of statements using the TextCNN model.

Background Knowledge Required

This course is for Huawei's basic certification. To better understand this course, familiarize yourself with the following:

- Basic knowledge of Python, MindSpore concepts, and Python programming.

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1 MobileNetV2 Image Classification

1.1 Introduction

In this exercise, the lightweight network MobileNetV2 is used to classify flower image datasets.

1.2 Preparations

Before you start, check whether MindSpore has been correctly installed. You are advised to install MindSpore on your computer by referring to the MindSpore official website https://www.mindspore.cn/install/en.

In addition, you should have basic mathematical knowledge, including knowledge of Python coding basics, probability, and matrix.

Recommended environment:

Version: MindSpore 1.7

Programming language: Python 3.7

1.3 Detailed Design and Implementation

1.3.1 Data Preparation

Image flower dataset used in the example is an open-source dataset and contains five flower types: daisies (633 images) dandelions (898 images), roses (641 images), sunflowers (699 images), and tulips (799 images). The 3670 photos, which are about 230 MB in total, are stored in five folders. To facilitate the model test after the model deployment, the dataset is divided into flower_photos_train and flower_photos_test.

The directory structure is as follows:

```
flower_photos_train
    ├── daisy
    ├── dandelion
    ├── roses
    ├── sunflowers
    ├── tulips
```

```
    ├── LICENSE.txt
flower_photos_test
    ├── daisy
    ├── dandelion
    ├── roses
    ├── sunflowers
    ├── tulips
    ├── LICENSE.txt
```

Obtain the datasets from the following links:

```
https://ascend-professional-construction-dataset.obs.myhuaweicloud.com/deep-learning/flower_photos_train.zip
https://ascend-professional-construction-dataset.obs.myhuaweicloud.com/deep-learning/flower_photos_test.zip
```

1.3.2 Procedure

Step 1 Load the dataset.

Define the create_dataset function, use the ImageFolderDataset API to load the flower image classification dataset, and perform image enhancement on the dataset. Code:

```
import mindspore.dataset as ds
import mindspore.dataset.vision.c_transforms as CV
from mindspore import dtype as mstype

train_data_path = 'flower_photos_train'
val_data_path = 'flower_photos_test'


def create_dataset(data_path, batch_size=18, training=True):
    """Define the dataset."""

    data_set = ds.ImageFolderDataset(data_path, num_parallel_workers=8, shuffle=True,
                                     class_indexing={'daisy': 0, 'dandelion': 1, 'roses': 2,
'sunflowers': 3,
                                                     'tulips': 4})
    # Perform image enhancement on the dataset.
    image_size = 224
    mean = [0.485 * 255, 0.456 * 255, 0.406 * 255]
    std = [0.229 * 255, 0.224 * 255, 0.225 * 255]
    if training:
        trans = [
            CV.RandomCropDecodeResize(image_size, scale=(0.08, 1.0), ratio=(0.75, 1.333)),
            CV.RandomHorizontalFlip(prob=0.5),
            CV.Normalize(mean=mean, std=std),
            CV.HWC2CHW()
        ]
    else:
        trans = [
```

```
                CV.Decode(),
                CV.Resize(256),
                CV.CenterCrop(image_size),
                CV.HWC2CHW()
            ]

    # Perform the data map, batch, and repeat operations.
    data_set = data_set.map(operations=trans, input_columns="image", num_parallel_workers=8)
    # Set the value of the batch_size. Discard the samples if the number of samples last fetched is
less than the value of batch_size.
    data_set = data_set.batch(batch_size, drop_remainder=True)

    return data_set


dataset_train = create_dataset(train_data_path)
dataset_val = create_dataset(val_data_path)
```

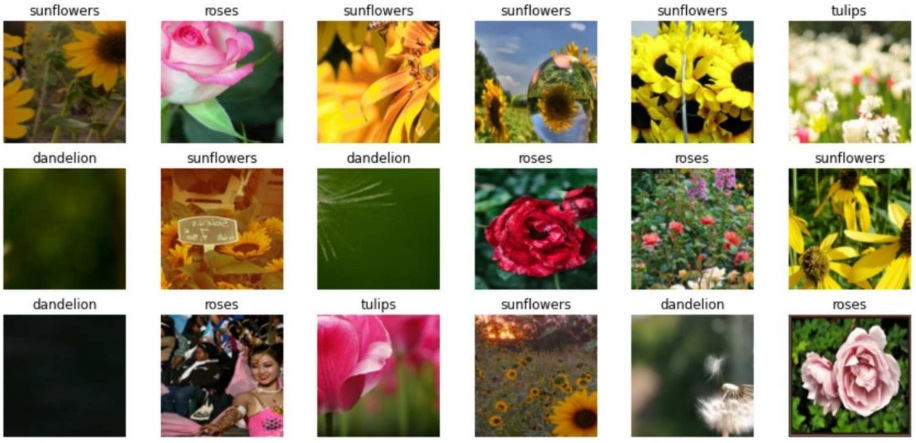
Step 2 Visualize the dataset.

The return value of the training dataset loaded from the create_dataset API is a dictionary. You can use the create_dict_iterator API to create a data iterator and use **next** to iteratively access the dataset. Here, **batch_size** is set to **18**. Therefore, you can use **next** to obtain 18 images and label data at a time. Code:

```
import matplotlib.pyplot as plt
import numpy as np

data = next(dataset_train.create_dict_iterator())
images = data["image"]
labels = data["label"]

print("Tensor of image", images.shape)
print("Labels:", labels)

# class_name corresponds to label. Labels are marked in ascending order of the folder character
 string.
class_name = {0:'daisy',1:'dandelion',2:'roses',3:'sunflowers',4:'tulips'}
plt.figure(figsize=(15, 7))
for i in range(len(labels)):
    # Obtain an image and its label.


    # Process images for display.
    data_image = np.transpose(data_image, (1, 2, 0))
    mean = np.array([0.485, 0.456, 0.406])
    std = np.array([0.229, 0.224, 0.225])
    data_image = std * data_image + mean
    data_image = np.clip(data_image, 0, 1)
    # Display the image.
```

```
    plt.title(class_name[int(labels[i].asnumpy())])
    plt.axis("off")
plt.show()
```

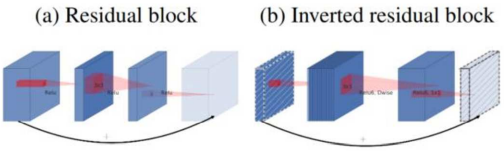
Output:



Step 3 Create a MobileNetV2 model.

Datasets are crucial for training. A good dataset can effectively improve training accuracy and efficiency. MobileNet is a lightweight CNN proposed by Google in 2017 to focus on mobile, embedded, and IoT devices. Compared with traditional convolutional neural networks, MobileNet uses depthwise separable convolution to greatly reduce the model parameters and computation amount with a slight decrease in accuracy. In addition, the width coefficient α and resolution coefficient β are introduced to meet the requirements of different application scenarios.

Because a large amount of data is lost when the ReLU activation function in the MobileNet processes low-dimensional feature information, the MobileNetV2 proposes to use an inverted residual block and Linear Bottlenecks to design the network, improving the accuracy of the model and making the optimized model smaller.



In the inverted residual block structure, the 1 x 1 convolution is used for dimension increase, the 3 x 3 depthwise convolution is used, and the 1 x 1 convolution is used for dimension reduction. This structure is opposite to the residual block structure. For the residual block, the 1 x 1 convolution is first used for dimension reduction, then the 3 x 3 convolution is used, and finally the 1 x 1 convolution is used for dimension increase.

For details, see the MobileNetV2 paper at https://arxiv.org/pdf/1801.04381.pdf.

Code:

```
import numpy as np
```

```
import mindspore as ms
import mindspore.nn as nn
import mindspore.ops as ops

def _make_divisible(v, divisor, min_value=None):
    if min_value is None:
        min_value = divisor
    new_v = max(min_value, int(v + divisor / 2) // divisor * divisor)
    # Make sure that round down does not go down by more than 10%.
    if new_v < 0.9 * v:
        new_v += divisor
    return new_v


class GlobalAvgPooling(nn.Cell):

    def __init__(self):
        super(GlobalAvgPooling, self).__init__()
        self.mean = ops.ReduceMean(keep_dims=False)

    def construct(self, x):
        x = self.mean(x, (2, 3))
        return x


class ConvBNReLU(nn.Cell):

    def __init__(self, in_planes, out_planes, kernel_size=3, stride=1, groups=1):
        super(ConvBNReLU, self).__init__()
        padding = (kernel_size - 1) // 2
        in_channels = in_planes
        out_channels = out_planes
        if groups == 1:
            conv = nn.Conv2d(in_channels, out_channels, kernel_size, stride, pad_mode='pad',
padding=padding)
        else:
            out_channels = in_planes
            conv = nn.Conv2d(in_channels, out_channels, kernel_size, stride, pad_mode='pad',
                             padding=padding, group=in_channels)

        layers = [conv, nn.BatchNorm2d(out_planes), nn.ReLU6()]
        self.features = nn.SequentialCell(layers)

    def construct(self, x):
        output = self.features(x)
        return output


class InvertedResidual(nn.Cell):

    def __init__(self, inp, oup, stride, expand_ratio):
        super(InvertedResidual, self).__init__()
        assert stride in [1, 2]

        hidden_dim = int(round(inp * expand_ratio))
```

```
        self.use_res_connect = stride == 1 and inp == oup

        layers = []
        if expand_ratio != 1:
            layers.append(ConvBNReLU(inp, hidden_dim, kernel_size=1))
        layers.extend([
            # dw
            ConvBNReLU(hidden_dim, hidden_dim,
                       stride=stride, groups=hidden_dim),
            # pw-linear
            nn.Conv2d(hidden_dim, oup, kernel_size=1,
                      stride=1, has_bias=False),
            nn.BatchNorm2d(oup),
        ])
        self.conv = nn.SequentialCell(layers)
        self.add = ops.Add()
        self.cast = ops.Cast()

    def construct(self, x):
        identity = x
        x = self.conv(x)
        if self.use_res_connect:
            return self.add(identity, x)
        return x


class MobileNetV2Backbone(nn.Cell):

    def __init__(self, width_mult=1., inverted_residual_setting=None, round_nearest=8,
                 input_channel=32, last_channel=1280):
        super(MobileNetV2Backbone, self).__init__()
        block = InvertedResidual
        # setting of inverted residual blocks
        self.cfgs = inverted_residual_setting
        if inverted_residual_setting is None:
            self.cfgs = [
                # t, c, n, s
                [1, 16, 1, 1],
                [6, 24, 2, 2],
                [6, 32, 3, 2],
                [6, 64, 4, 2],
                [6, 96, 3, 1],
                [6, 160, 3, 2],
                [6, 320, 1, 1],
            ]

        # building first layer
        input_channel = _make_divisible(input_channel * width_mult, round_nearest)
        self.out_channels = _make_divisible(last_channel * max(1.0, width_mult), round_nearest)
        features = [ConvBNReLU(3, input_channel, stride=2)]
        # building inverted residual blocks
        for t, c, n, s in self.cfgs:
            output_channel = _make_divisible(c * width_mult, round_nearest)
            for i in range(n):
                stride = s if i == 0 else 1
```

```
                features.append(block(input_channel, output_channel, stride, expand_ratio=t))
                input_channel = output_channel
        # building last several layers
        features.append(ConvBNReLU(input_channel, self.out_channels, kernel_size=1))
        # make it nn.CellList
        self.features = nn.SequentialCell(features)
        self._initialize_weights()

    def construct(self, x):
        x = self.features(x)
        return x

    def _initialize_weights(self):

        self.init_parameters_data()
        for _, m in self.cells_and_names():
            if isinstance(m, nn.Conv2d):
                n = m.kernel_size[0] * m.kernel_size[1] * m.out_channels
                m.weight.set_data(ms.Tensor(np.random.normal(0, np.sqrt(2. / n),

m.weight.data.shape).astype("float32")))

                if m.bias is not None:
                    m.bias.set_data(
                        ms.numpy.zeros(m.bias.data.shape, dtype="float32"))
            elif isinstance(m, nn.BatchNorm2d):
                m.gamma.set_data(
                    ms.Tensor(np.ones(m.gamma.data.shape, dtype="float32")))
                m.beta.set_data(
                    ms.numpy.zeros(m.beta.data.shape, dtype="float32"))

    @property
    def get_features(self):
        return self.features


class MobileNetV2Head(nn.Cell):

    def __init__(self, input_channel=1280, num_classes=1000, has_dropout=False, activation="None"):
        super(MobileNetV2Head, self).__init__()
        # mobilenet head
        head = ([GlobalAvgPooling()] if not has_dropout else
                [GlobalAvgPooling(), nn.Dropout(0.2)])
        self.head = nn.SequentialCell(head)
        self.dense = nn.Dense(input_channel, num_classes, has_bias=True)
        self.need_activation = True
        if activation == "Sigmoid":
            # set activation

        elif activation == "Softmax":
            # set activation

        else:
            self.need_activation = False
        self._initialize_weights()
```

```
    def construct(self, x):
        x = self.head(x)
        x = self.dense(x)
        if self.need_activation:
            x = self.activation(x)
        return x

    def _initialize_weights(self):

        self.init_parameters_data()
        for _, m in self.cells_and_names():
            if isinstance(m, nn.Dense):
                m.weight.set_data(ms.Tensor(np.random.normal(
                    0, 0.01, m.weight.data.shape).astype("float32")))
                if m.bias is not None:
                    m.bias.set_data(
                        ms.numpy.zeros(m.bias.data.shape, dtype="float32"))

class MobileNetV2Combine(nn.Cell):

    def __init__(self, backbone, head):
        super(MobileNetV2Combine, self).__init__(auto_prefix=False)
        self.backbone = backbone
        self.head = head

    def construct(self, x):
        x = self.backbone(x)
        x = self.head(x)
        return x

def mobilenet_v2(num_classes):
    backbone_net = MobileNetV2Backbone()
    head_net = MobileNetV2Head(backbone_net.out_channels,num_classes)
    return MobileNetV2Combine(backbone_net, head_net)
```

Step 4 Train and validate the model.

After a model, a loss function and an optimizer are created, the Model API is used to initialize the model, the model.train API is used to train the model, and the model.eval API is used to validate the model accuracy.

This section involves the following knowledge of transfer learning:

1. Download a pre-trained model weight.

   Download the model file pre-trained on the ImageNet dataset, and save it to the directory at the same level as the running code. Download link: https://download.mindspore.cn/models/r1.7/mobilenetv2_ascend_v170_imagenet2012_official_cv_top1acc71.88.ckpt.

2. Read the pre-trained model.

   Read the pre-trained model file through the load_checkpoint() API. The output result is in dictionary data format.

3. Modify pre-trained model parameters.

Modify the parameters related to the pre-trained model weight. (The model is pre-trained on the ImageNet dataset to classify 1001 types. However, the current exercise is to classify five types of flowers. The network model modifies the last fully-connected layer.)

Code:

```
import mindspore
import mindspore.nn as nn
from mindspore.train import Model
from mindspore import Tensor, save_checkpoint
from mindspore.train.callback import ModelCheckpoint, CheckpointConfig, LossMonitor
from mindspore.train.serialization import load_checkpoint, load_param_into_net
# Create a model. The number of target classes is 5.
network = mobilenet_v2(5)

# Load the pre-trained weight.
param_dict =
load_checkpoint("./mobilenetv2_ascend_v170_imagenet2012_official_cv_top1acc71.88.ckpt")

# Modify the weight data based on the modified model structure.
param_dict["dense.weight"] =
mindspore.Parameter(Tensor(param_dict["dense.weight"][:5, :],mindspore.float32),
name="dense.weight", requires_grad=True)
param_dict["dense.bias"] =
mindspore.Parameter(Tensor(param_dict["dense.bias"][:5, ],mindspore.float32), name="dense.bias",
requires_grad=True)

# Load the modified weight parameters to the model.
load_param_into_net(network, param_dict)

train_step_size = dataset_train.get_dataset_size()
epoch_size = 20
lr = nn.cosine_decay_lr(min_lr=0.0, max_lr=0.1,total_step=epoch_size *
train_step_size,step_per_epoch=train_step_size,decay_epoch=200)
# Define the optimizer.
network_opt = nn.Momentum(params=network.trainable_params(), learning_rate=0.01,
momentum=0.9)

# Define the loss function.


# Define evaluation metrics.


# Initialize the model.


# Monitor the loss value.


# Set the number of steps for saving a model and the maximum number of models that can be
saved.
```

```
# Save the model. Set the name, path, and parameters for saving the model.


print("============== Starting Training ==============")
# Train a model, set the number of training times to 5, and set the training set and callback function.
model.train(5, dataset_train, callbacks=[loss_cb,ckpoint_cb], dataset_sink_mode=True)
# Use the test set to validate the model and output the accuracy of the test set.
metric = model.eval(dataset_val)
print(metric)
```

Output:

```
============== Starting Training ==============
epoch: 1 step: 201, loss is 0.8389087915420532
epoch: 2 step: 201, loss is 0.5519619584083557
epoch: 3 step: 201, loss is 0.26490363478660583
epoch: 4 step: 201, loss is 0.4540162682533264
epoch: 5 step: 201, loss is 0.5963617563247681
{'Accuracy': 0.9166666666666666}
```

Step 5 Visualize the model prediction result.

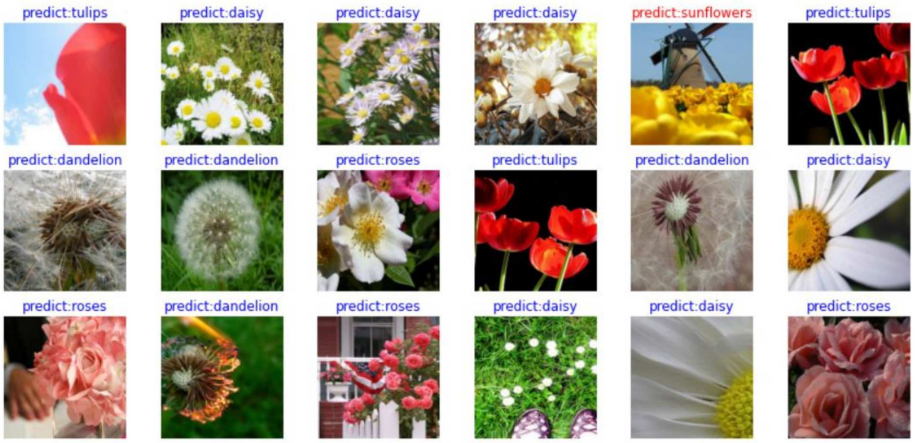
Define the visualize_model function, use the model with the highest validation accuracy described above to predict the input image and visualize the prediction result.

Code:

```
import matplotlib.pyplot as plt
import mindspore as ms

def visualize_model(best_ckpt_path, val_ds):
    num_class = 5  # Perform binary classification on wolf and dog images.
    net = mobilenet_v2(num_class)
    # Load model parameters.




    # Load the validation dataset.
    data = next(val_ds.create_dict_iterator())
    images = data["image"].asnumpy()
    labels = data["label"].asnumpy()
    class_name = {0:'daisy',1:'dandelion',2:'roses',3:'sunflowers',4:'tulips'}
    # Predict the image type.
    output = model.predict(ms.Tensor(data['image']))
    pred = np.argmax(output.asnumpy(), axis=1)

    # Display the image and the predicted value of the image.
    plt.figure(figsize=(15, 7))
    for i in range(len(labels)):
        plt.subplot(3, 6, i + 1)
        # If the prediction is correct, it is displayed in blue. If the prediction is incorrect, it is di
splayed in red.
        color = 'blue' if pred[i] == labels[i] else 'red'
        plt.title('predict:{}'.format(class_name[pred[i]]), color=color)
```

```
        picture_show = np.transpose(images[i], (1, 2, 0))
        mean = np.array([0.485, 0.456, 0.406])
        std = np.array([0.229, 0.224, 0.225])
        picture_show = std * picture_show + mean
        picture_show = np.clip(picture_show, 0, 1)
        plt.imshow(picture_show)
        plt.axis('off')

    plt.show()

visualize_model('ckpt/mobilenet_v2-5_201.ckpt', dataset_val)
```

Output:



1.4 Question

What API is used to read pre-trained models?